Twitter Sentiment Analysis

of Tech Products

NLP Pipeline · Multi-Model Evaluation · Interactive Dashboard

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Problem & Business Value

The Problem

1. Consumer brands receive thousands of tweets daily about their products
2. Manual sentiment monitoring is impossible at scale
3. A single viral negative thread can precede measurable sales decline

Our Solution

TARGET

19K+ Apple & Google tweets

APPROACH

5 NLP models compared rigorously

METRIC

Negative Recall (catch every complaint)

OUTPUT

Interactive real-time dashboard

Primary Metric: Negative Recall — missed negative signals cost more than false positives

Dataset Overview

19,038	41.8%	29.2%	28.2%
Total Tweets	Positive	Negative	Neutral

Source Datasets

File	Source	Size	Labels
Provided.csv	Crowdflower	~9,000	3-class
Data2.csv	Sentiment140	~10,000	Binary

Key Statistics

Avg. Tweet Length	**74 characters**
Avg. Word Count	**13.2 words**
Train / Test Split	**80% / 20% stratified**
Deduplication	**Case-insensitive cross-dataset**
Brand Focus	**Apple & Google products**

NLP Preprocessing Pipeline

The same pipeline runs at training time and in the live dashboard — ensuring zero train/serve skew

1 **Lowercase**
Normalise casing

2 **Remove URLs & @mentions**
Strip noise tokens

3 **Expand #hashtags**
Keep semantic content

4 **Strip special chars**
Letters & spaces only

5 **Tokenise (NLTK)**
word_tokenize()

6 **Remove stopwords**
KEEP: not, no, never, n't

⭐ **Key design decision:** Negation words (not, no, never, n't) are deliberately kept — they reverse sentiment polarity and are critical for negative recall.

Models Evaluated

VADER

Rule-Based Baseline

FEATURES
Lexicon

TASK
3-class

Logistic Regression

Linear Multiclass

FEATURES
TF-IDF

TASK
3-class

Naive Bayes

Binary Classifier

FEATURES
CountVectorizer

TASK
Pos vs Neg

LightGBM

Leaf-Wise Boosting

FEATURES
TF-IDF

TASK
3-class

XGBoost

RECOMMENDED

Level-Wise Boosting

FEATURES
TF-IDF

TASK
3-class

All ML models evaluated on a shared 20% stratified held-out test set · 5-fold cross-validation for LR and XGBoost

Model Results

Primary: Negative Recall · Secondary: Macro F1 · All measured on identical 20% hold-out set

#	Model	Neg Recall	Macro F1	Task
1	Naive Bayes (Binary)	0.835	0.743	Binary
2	XGBoost ⭐	0.804	0.672	3-class
3	Logistic Regression	0.750	0.711	3-class
4	LightGBM	0.740	0.685	3-class
5	VADER (Baseline)	0.427	0.519	3-class

Key Insights

🥇 **Naive Bayes — highest recall**

0.82 Neg Recall — but only on binary (Pos/Neg). Cannot distinguish Neutral.

⭐ **XGBoost — production pick**

Handles all 3 classes. Best Macro F1 (0.74). Exceeds the 0.70 Neg Recall target.

📉 **VADER — weak baseline**

Misses 61% of negatives. Cannot learn domain jargon or sarcasm.

🎯 **Target exceeded**

3 of 4 ML models beat the 0.70 Negative Recall business threshold.

Interactive Streamlit Dashboard

Analyze Tweet

Single-tweet prediction with confidence scores and a radar chart breakdown per sentiment class. Quick-example buttons pre-load demo tweets.

Batch Analysis

Upload a CSV (or paste tweets directly). Outputs sentiment distribution pie chart, confidence histogram, results table, and a downloadable CSV.

Model Comparison

Interactive leaderboard showing all 5 models ranked by Neg Recall and Macro F1. Grouped bar chart with 0.70 reference line.

Insights

Top-word frequency charts per sentiment class. Dataset statistics panel. Step-by-step preprocessing pipeline visualisation.

All models selectable at runtime · Twitter dark theme UI · Identical preprocessing between training and serving · streamlit run streamlit_app.py

Conclusions & Recommendations

Deploy XGBoost

Best balance of Negative Recall and Macro F1 across all three sentiment classes. Production-ready.

Use Naive Bayes for speed

If latency is critical and Neutral class isn't required, Naive Bayes delivers the highest Negative Recall with the fastest inference.

Refresh training data

Twitter language evolves fast. Re-train quarterly on fresh Apple/Google tweets to maintain signal quality.

Preserve negation

Never strip negation words from preprocessing. These tokens are the primary driver of negative recall gains.

Validation Strategy

Can our models actually handle tweets they have never seen before?

Stratified 5-Fold Cross-Validation

Fold 1	Fold 2	Fold 3	Fold 4	**TEST**

← *Repeats 5×, rotating the TEST fold each time*

Final score = **mean ± std deviation** across all 5 test folds

What We Measure

Metric	Why it matters
Negative Recall	Primary — catches every complaint; low score = complaints slip through undetected
Macro F1	Secondary — balanced across all 3 classes, not just the majority

The Overfitting Red Flag

Train score − CV score > 0.05
means the model memorised training noise and will underperform in production. We flag any model exceeding this gap.

Models Under Validation

- ✅ **Logistic Regression**
- ✅ **LightGBM**
- ✅ **XGBoost**
- ⛔ VADER *(rule-based, no training)*
- ⛔ Naive Bayes *(binary only)*

Thank You

Built with curiosity, cleaned with care, deployed with pride.

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🔗 **Live Dashboard**

STREAMLIT UI

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